Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/24/2004
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Project Title:	Optical Biosensors for In-Flight Measurement of Hormone Levels		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smith, Scott (NASA Johnson Space Center)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	As part of University of California at Los Angeles (UCLA) and NASA-Johnson Space Center (JSC)'s effort to develop a bioassay system for in-flight measurement of hormone levels, we will deliver to NASA a fiber optic unit. The fiber-optic biosensor is based on a sol-gel immobilization technique where antibodies specific for the target of interest are immobilized in a solid-state material. The innovative feature of this biosensor is that we can bring well established immunochemistry analytical techniques into the realm of solid state materials. By coupling the biomaterial to a fiber-optic system, we can use simple instrumentation with a quick response time. The resulting fiber-optic biosensor exploits the advantageous aspects of each field: the immunochemistry provides high specificity and sensitivity for the hormone of interest and accurate methods of quantification while the solid-state element is small and compact, facilitating space crew handling and a simple procedure. The principal objective for our research this year was to develop the analytical technique for thin-film immunosensors, a vital step in preparation for fiber-optic work. A secondary objective was to compare cortisol measurements made using		

Task Description:	the sol-gel immunosensors with those measured at JSC using their standard techniques. Finally, development of the fiber-optic immunosensor was to be initiated. During this reporting period, we made excellent progress in the area of thin-film development, and in verifying our data with that of NASA-JSC. A calibration curve for the fiber optic device was obtained. The objective of this task is to develop sol-gel based biosensor systems for biological analytes for in-flight use. The candidates are hormones such as cortisol and PTH (parathyroid hormone) and collagen breakdown products such as pyridinium crosslinks (pyridinoline and deoxypyridinoline). The sensor systems will have the sensitivity and accuracy for detecting the analyte levels in physiological range. They will also have storage stability and be compact and easy to handle for usage in space. The sensor systems will be sol-gel based immunosensors that offer several advantages for use in space. The sensor material will consist of antibodies stabilized in a rugged, solid-state environment, which will avoid excess liquid transfer steps during an assay and provide easier handling in space. The sensor material will be coupled to fiber optic system, in which signals feed directly into a spectrophotometer for fast detection. Such systems will be truly compact and convenient for use in the limited space available in a spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research and development of new biomaterials will have applicability to a wide range of biosensors. The biologically active materials we are developing offer solid-state sensing elements that are more rugged and stable for the biotechnology community. The technical advances associated with this research can be adapted to detection of a wide range of analytes.
Task Progress:	Over the past year, there have been several improvements in the sol-gel optical biosensor for cortisol. The first has been to improve the processing and optical quality of the silica sol-gel matrix. We have begun to use a sodium phosphate buffer of a lower pH to obtain a consistent gelation time of about 30 minutes for dip-coating sol-gel thin films. This longer gelation time allows for facile pulling of thin films. Also, preparation of the sol prior to pulling films has enabled good mixing of the sol constituents, leading to smooth, uniform films. Previously, our methods created bubbles, which added roughness to the thin films and variable optical properties. The second area of study was in the development of the immunoassay, specifically using a new analytical procedure to analyze luminescence data to more accurately correlate results with those obtained from NASA-JSC. Well-defined calibration curves have been obtained with the new sol-gel matrix. The use of dichloromethane for the extraction of steroids from JSC serum was investigated. While the results were promising, we decided against further development because the use of dichloromethane in space precluded practical use of this method in-flight. A second approach we investigated was to essentially make our own standard serum solutions out of JSC serum with known cortisol concentrations for the standards. Unfortunately, we were unsuccessful in being able to correlate results from actual JSC serum with the calibration curve obtained from the standards we made. The major problem is that there is a significant difference in the background noise from other biomolecules in the serum and that the noise varies from sample to sample, so that we are unable to obtain a good correlation for several JSC serum samples with one standard curve. A new analytical method has been developed that does not require dichloromethane extraction of steroids from serum. In this approach, one untreated serum sample is divided into three samples of equal volume (e.g., 200 microliters) and a certain volume of cortisol solution is added such that we know the deviation in concentration from the initial unknown cortisol concentration. Fluorescence measurements are taken using these “spiked” serum samples in our immunoassay. The relationship of fluorescence varying linearly with the logarithm of concentration has been established so three separate equations can be written for each “spiked” serum sample. Since there are three unknown variables (slope of the curve, y-intercept, and unknown cortisol concentration) and three independent equations, the cortisol concentration can be calculated. The advantages of this method for analyzing serum samples are that it does not require steroid extraction with a toxic compound and that it analyzes samples individually so we don’t have to deal with the disparity of different JSC serum samples. With this method, we have obtained cortisol concentration values that have been in complete agreement with the measurements obtained by NASA-JSC through traditional methods. The third area of investigation was in the development of fiber-optic immunosensor. During the year, the first experiments with optical fibers have been completed. To prepare the optical fibers for sol-gel coating, approximately 3 cm of the buffer coating on the optical fibers were stripped off using boiling sulfuric acid and the bare fibers were stored in water. Then, the bare optical fiber was immersed into piranha solution (70% sulfuric acid/30% hydrogen peroxide) for at least several hours before being rinsed with water. This cleaning procedure enhances the adherence of the sol-gel coating to the glass substrate, whether it is a cover slip or a fiber. The immunoassay protocol for the sol-gel antibody coated optical fibers was analogous to that with thin films in that the fibers were dipped into test tubes containing a solution with labeled cortisol and serum. Using this approach, we have obtained a calibration curve of fluorescence vs. cortisol concentration with the fiber-optic system.
Bibliography Type:	Description: (Last Updated: 02/11/2020)
Articles in Peer-reviewed Journals	Zhou JC, Chuang MH, Lan EH, Dunn B, Gillman PL, Smith SM. "Immunoassays for cortisol using antibody-doped sol-gel silica." J Mater Chem. 2004;14(14):2311-6. https://doi.org/10.1039/b401902d , Jul-2004